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NSAIDs, leukotriene antagonists, prostanoid derivatives

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PYCH304 (Pharmacology I)
17th September 2025 (8.00-10.00)

Learning Objectives

After the class, students should be able to

1. **Explain the mechanism of action (MOA), pharmacokinetic and pharmacodynamic properties of NSAIDs**
2. **Explain ADRs and drug interactions of NSAIDs**
3. Describe the mechanism of action, indication(s), and ADRs of anti-leukotrienes
4. Recognize the indication(s), MOA, and ADRs of prostanoid derivatives

An overview of autacoids

- **Autacoids** are substances that are
  - Rapidly synthesized in response to specific stimuli
  - Act quickly in the immediate environment,
  - Remain active for only a short time before degradation

- **Major classes of autacoids**
  1. *Biogenic amines:* Histamine, serotonin (5-hydroxytryptamine)
  2. *Polypeptides:* Bradykinin, angiotensin
  3. *Lipid-derived autacoids*
     - *Eicosanoids:* Prostaglandins, leukotrienes, thromboxane
     - *Platelet activating factor (PAF)*

Introduction

- **Eicosanoids ("*eikosi*" *is Greek term* = twenty)**
  - Lipid-derived autacoids that contain 20 carbons
  - Mostly derived from arachidonic acid

- Drugs affecting eicosanoid pathways include
  1. **Non-steroidal anti-inflammatory drugs (NSAIDs)**
  2. **Leukotriene inhibitors**
  3. **Prostaglandin analogues**

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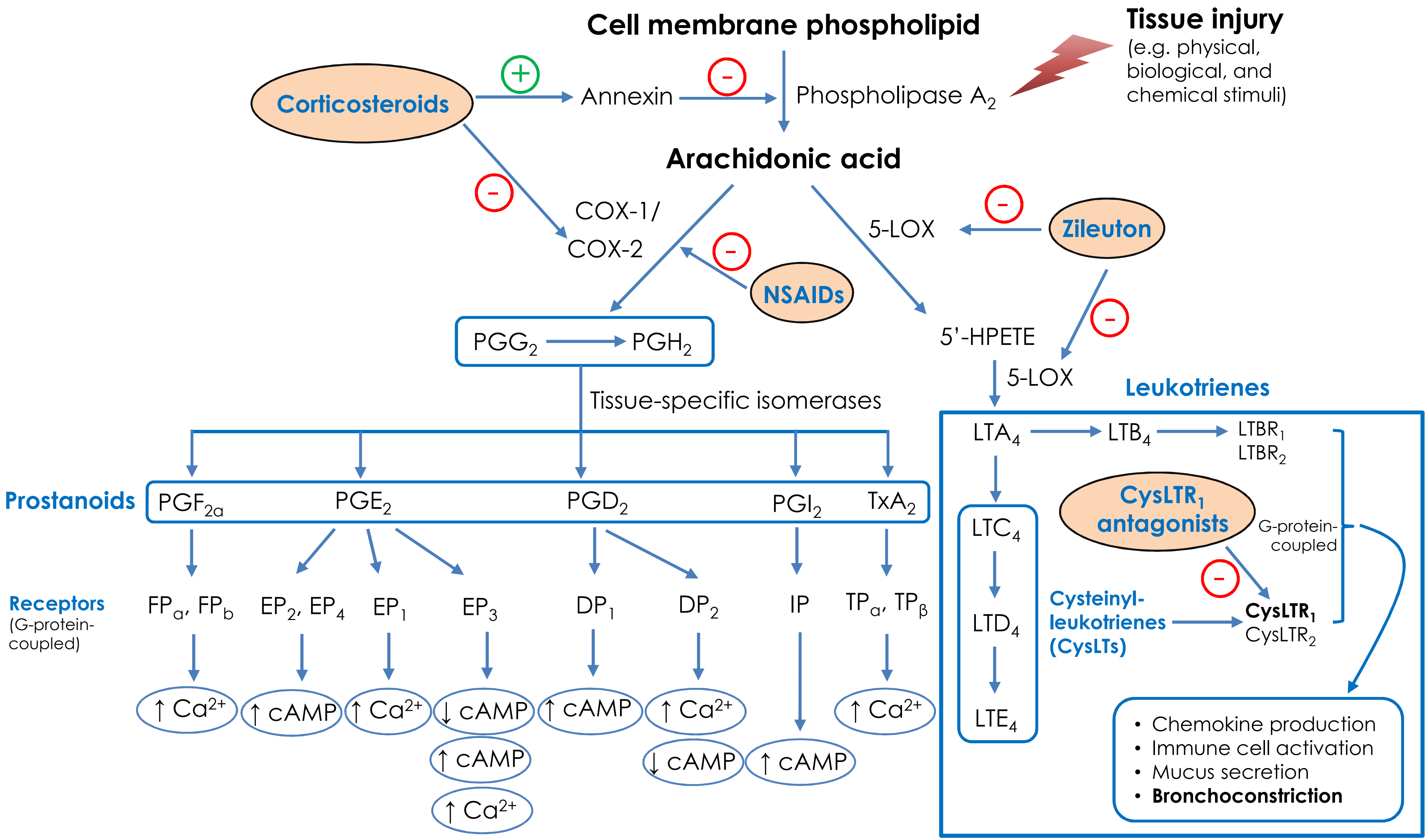
Eicosanoid Synthesis Pathways



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Characteristics of COX Isozymes

COX isozymes	Characteristics
COX-1	• **Constitutively expressed** in most cell types • Regulates several homeostatic processes (housekeeping)
COX-2	• Present in many cells (e.g. endothelial cells, macrophages, synovial fibroblasts, mast cells, chondrocytes and osteoblasts) • **Induced by cytokines and other inflammatory stimuli**
COX-3	• A splice variant of COX-1 that retains the entire COX-1 transcript • **Mainly expressed in the brain and spinal cord** and to a lesser extent in the heart, endothelial cells and monocytes • May have a role in pain perception • **Also sensitive to acetaminophen**

Curr Pharm Des. 2003;9(27):2177-90.

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Biological Effects of Prostanoids

Organ/tissue	Prostanoids	Effects
Female Reproductive Organs	PGE_2, PGF_{2a}	Uterine Contraction, Oxytocic Action
Male Reproductive Organs	PGE_2, PGF_{2a}	Fertility, Penile erection
Cardiovascular System	TXA_2, PGI_2 TXA_2 PGE_2, PGI_2 TXA_2, PGF_{2a} PGE_2, PGI_2	Thrombosis, Platelet Aggregation Vascular Permeability Arterial Vasodilation Venous vasoconstriction Patency of the Fetal Ductus Arteriosus
Respiratory System	PGE_2 PGF_{2a}, TXA_2	Bronchodilation Bronchoconstriction
Renal System	PGE_2, PGI_2 PGE_2, PGI_2 PGE_2	Regulation Renal Blood Flow and GFR Renin Release Inhibition Hydroosmotic Effect of ADH
Gastrointestinal System	PGE_2, PGI_2	Cytoprotection
Immune System	PGE_2, PGI_2	Inhibit T & B cells activation & proliferation
Central Nervous System	PGE_2 PGD_2 PGE_2, PGI_2	Fever Sleep Pain
Ocular tissue	PGD_2, PGE_2, PGF_{2a}	Control the drainage of aqueous humor out of the eye

[1]Curr Pharm Des. 2003;9(27):2177-90. [2]Clin Lipidology. 2011;6(5):577-591.

Pharmacology of Non-steroidal Anti-inflammatory Drugs

- Prostanoid biosynthesis is increased in inflamed tissue
  - As pro-inflammatory mediators (primarily PGE_2 and PGI_2)
    - ↑ Local blood flow, vascular permeability, leukocyte infiltration

- **NSAIDs competitively/reversibly inhibit COX enzymes**
  - **Except Aspirin: irreversible COX inhibition**

⬇

↓ PG synthesis

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Structural Classification of NSAIDs

NSAIDs

- **Salicylic â derivatives**
  - Aspirin
  - Diflunisal
  - Sulfasalazine
- **Acetic â derivatives**
  - **Indole & Indene acetic â**
    - Indomethacin
    - Sulindac
    - Etodolac
  - **Heteroaryl acetic â**
    - Diclofenac
    - Ketorolac
    - Tolmetin
- **Arylpropionic â**
  - Ibuprofen
  - Flurbiprofen
  - Ketoprofen
  - Fenprofen
  - Naproxen
  - Oxaprozin
- **Fenamates (Anthranilic â)**
  - Mefenamic â
  - Eclofenamic â
- **Enolic â**
  - **Oxicams**
    - Piroxicam
    - Tenoxicam
  - **Pyrazolidones**
    - Phenylbutazone
    - Oxyphenthatrazone
- **Alkanones**
  - Nabumetone
- **Diarylheterocycles (COX-2 selective inhibitors)**
  - Rofecoxib
  - Celecoxib
  - Valdecoxib
  - Parecoxib — An injectable prodrug of valdecoxib
  - Etoricoxib
  - Lumiracoxib

Circulation. 2007;115:1634-1642.

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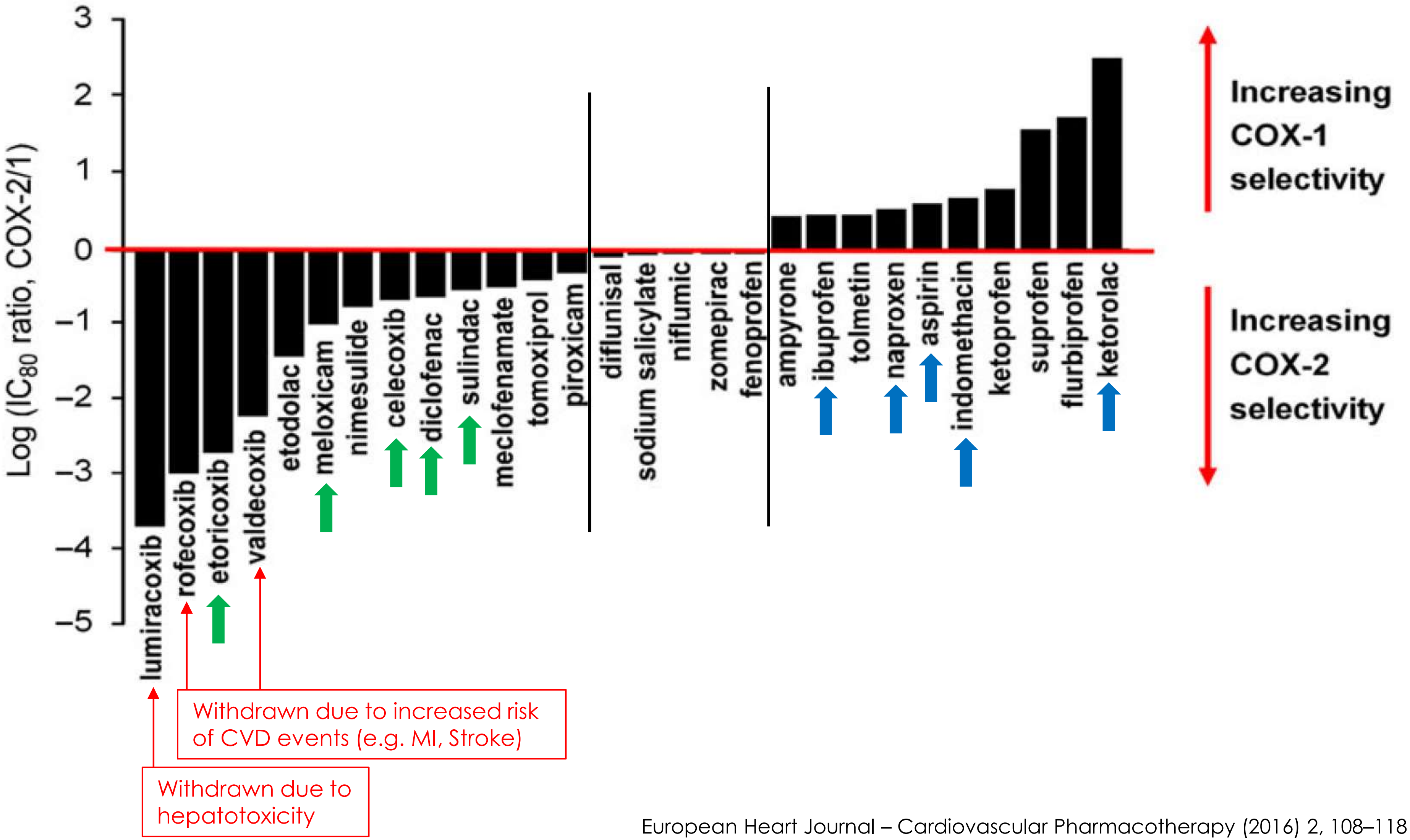
COX-2 Selectivity of NSAIDs

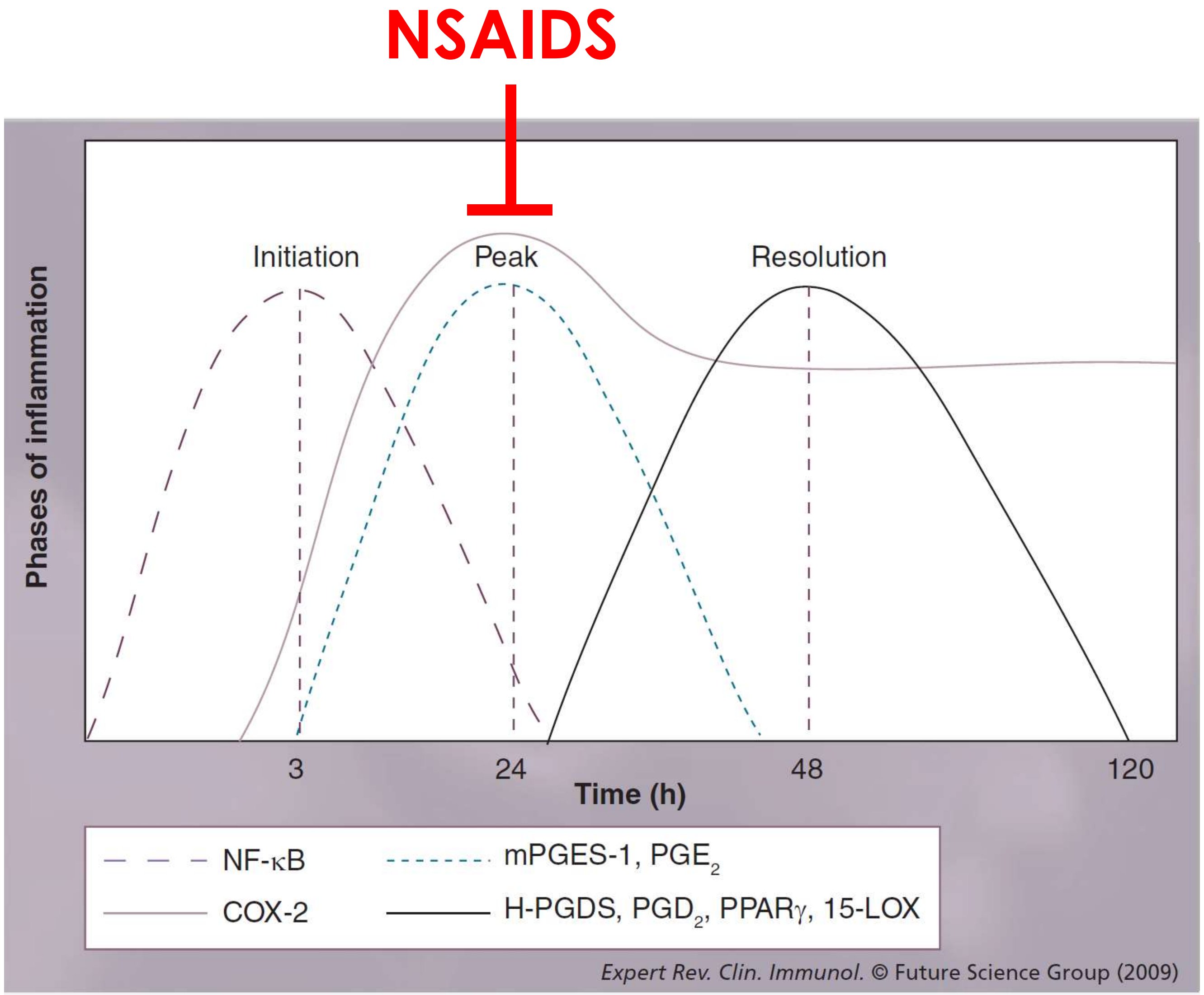


European Heart Journal – Cardiovascular Pharmacotherapy (2016) 2, 108–118

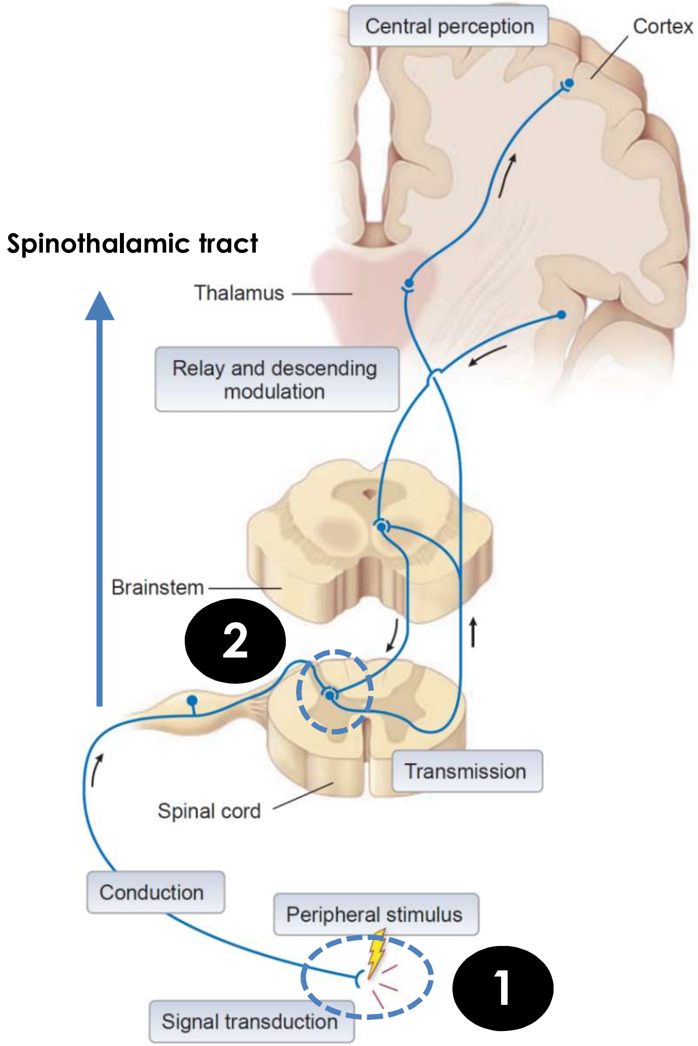
Pharmacological Effects of NSAIDs

- **Anti-inflammatory**
- **Analgesic**
- **Antipyretic**
- **Antiplatelet**
- **Tocolytic**
- **Induce closure of ductus arteriosus**

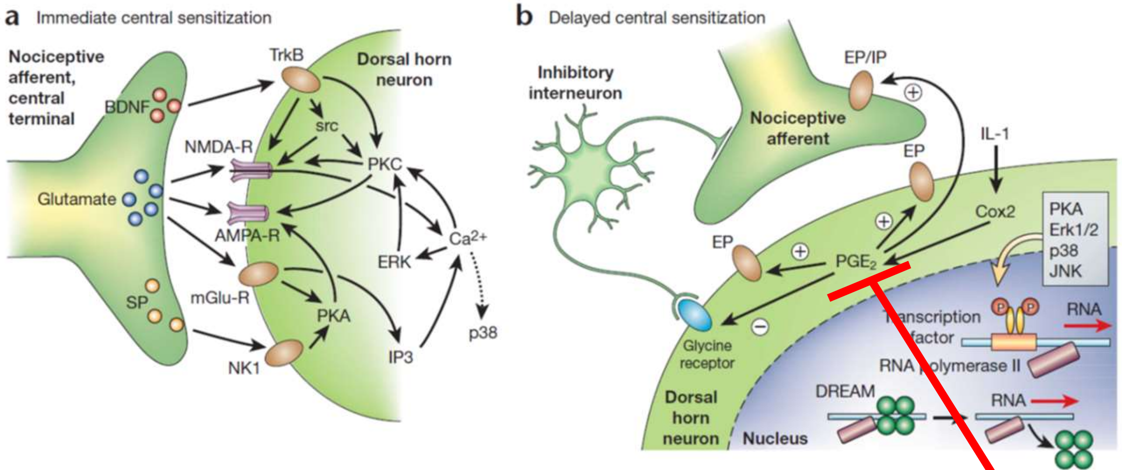
Anti-inflammatory Effect of NSAIDs



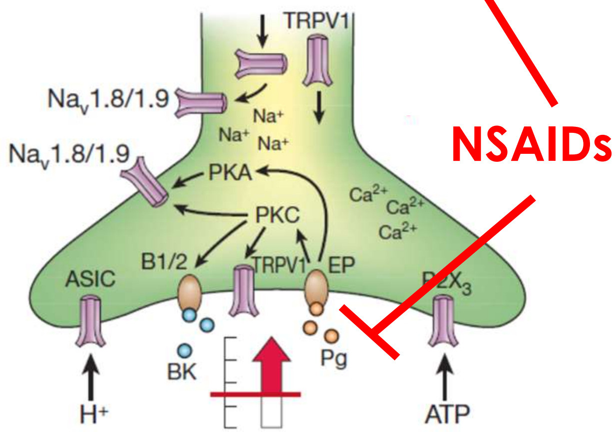
Analgesic Effects of NSAIDs



2 Dorsal horn neuron of spinal cord



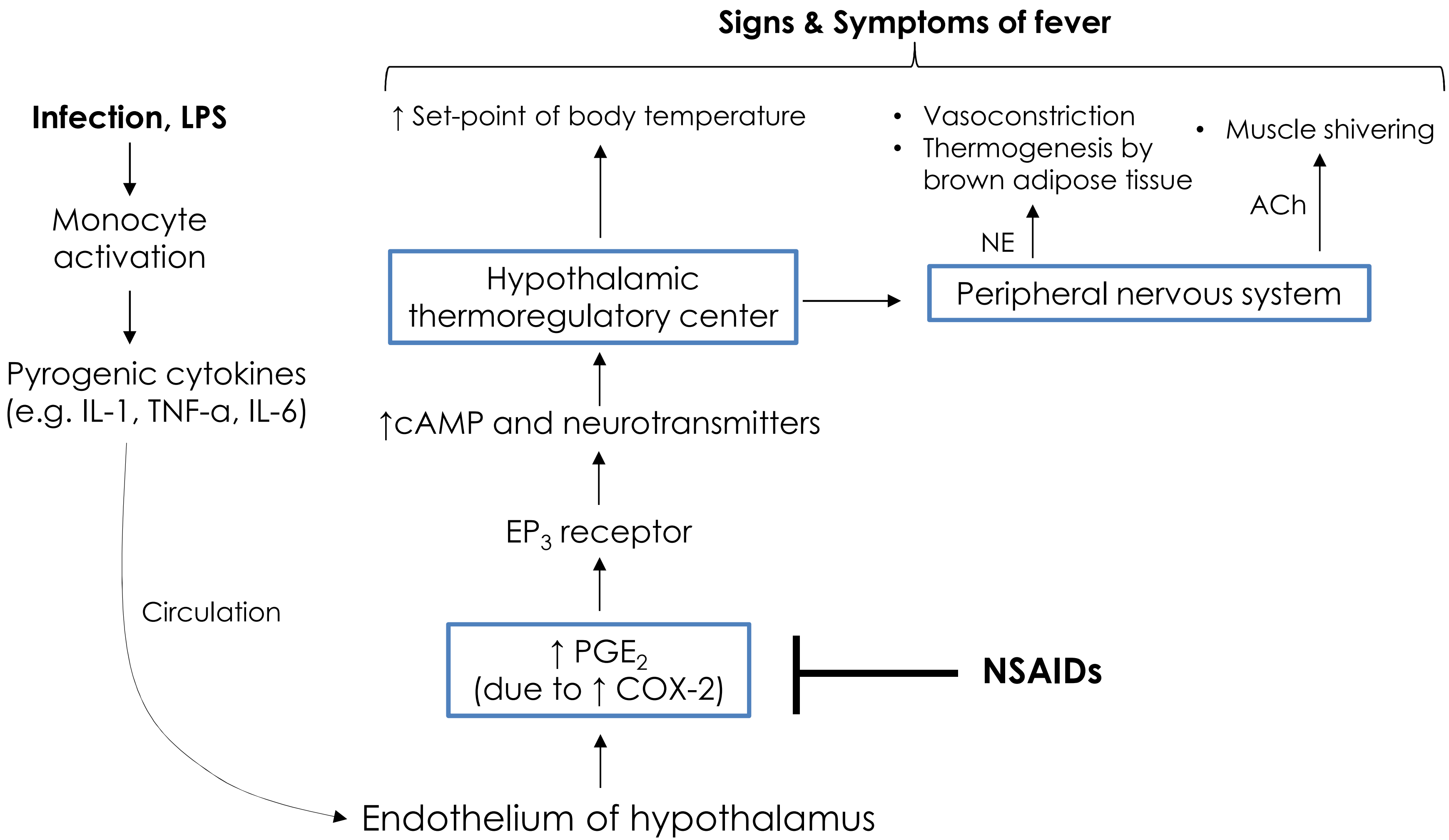
1 Peripheral sensory neuron (Aδ-fiber or C-fiber)



Principles of pharmacology : the pathophysiologic basis of drug therapy, 3rd ed. 2012.

Nat Neurosci. 2002;5 Suppl:1062-7.

Antipyretic Effects of NSAIDs



[1]Current Biology. 1999;9:R147–R150. [2]Nat Rev Immunol. 2015;15(6):335-49.

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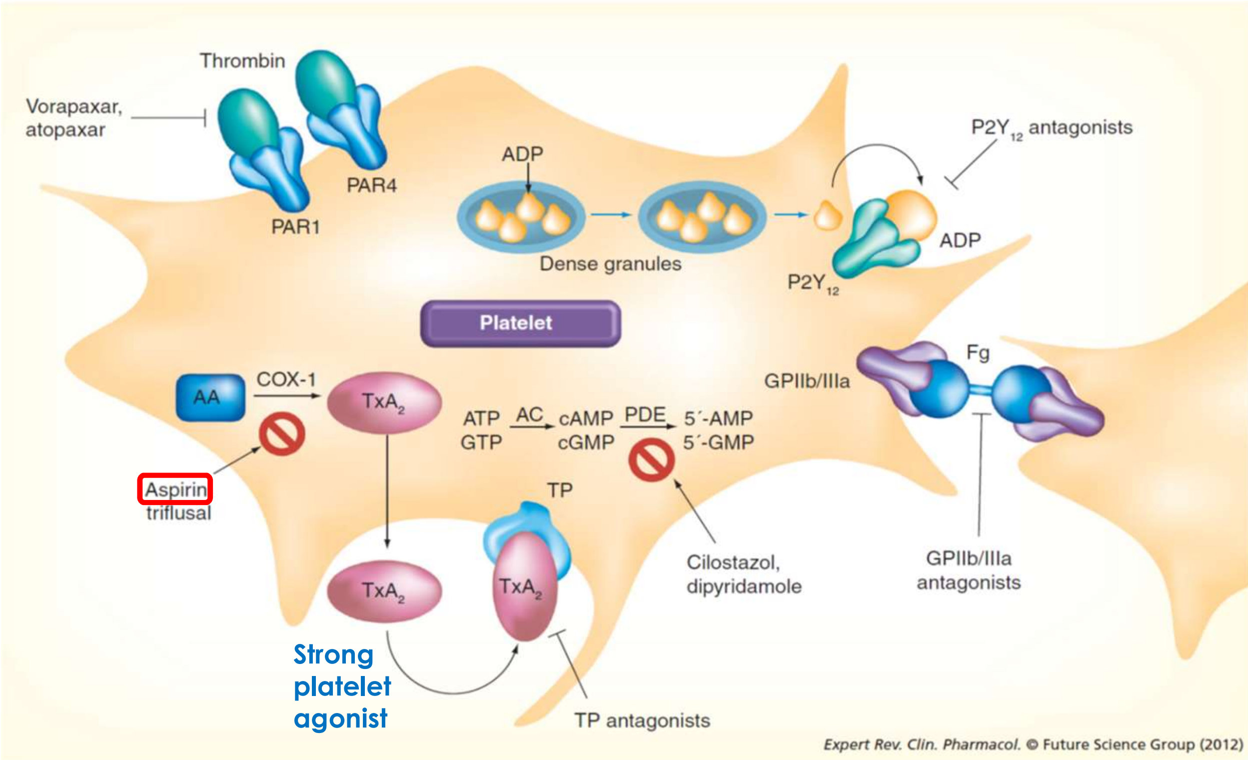
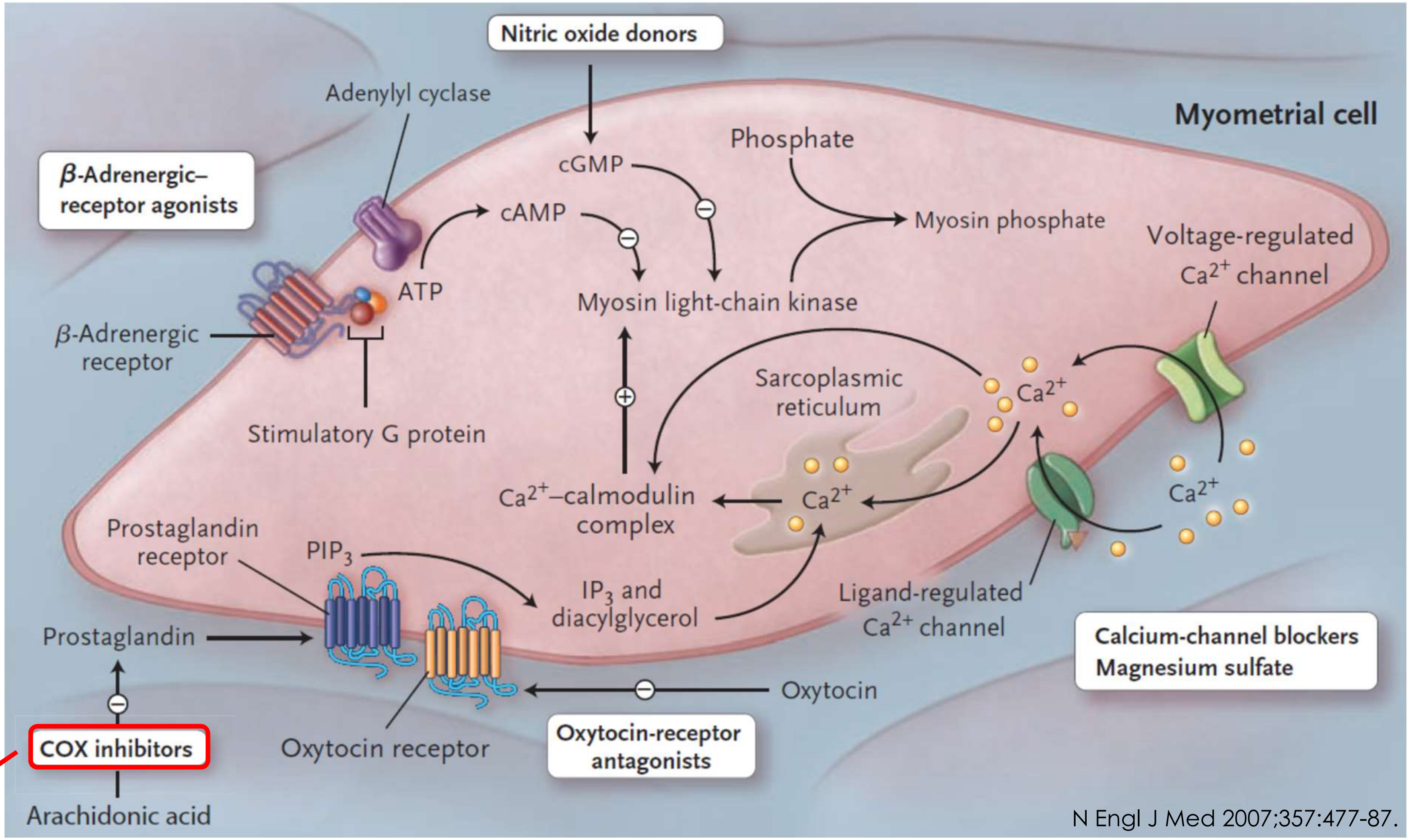
Antiplatelet Effect of NSAIDs



- **Low-dose aspirin**
  - Begins to acetylate platelets within minutes of reaching the presystemic circulation
  - Commonly used as antiplatelet agent for the prevention and treatment of thromboembolic events

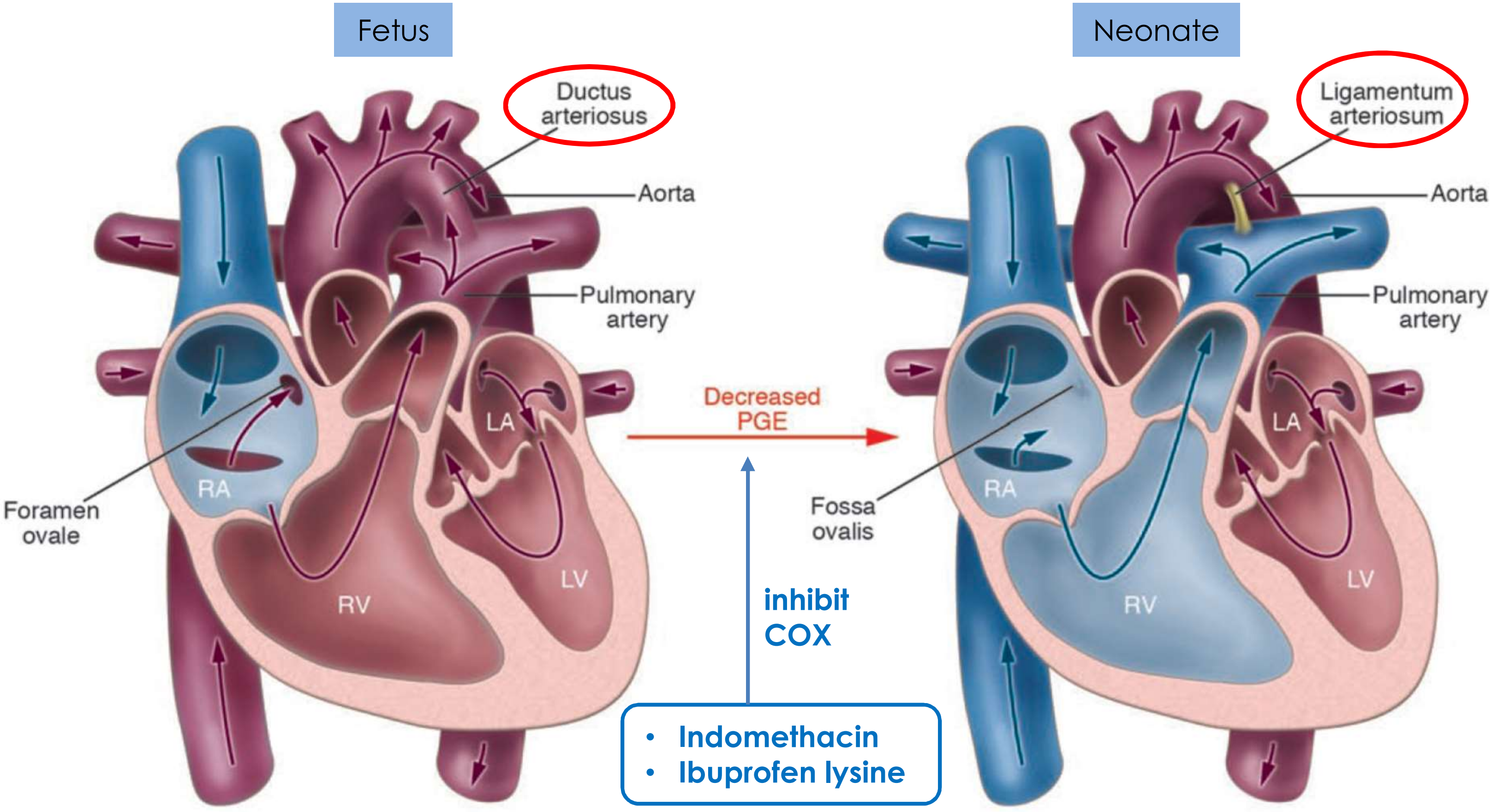
Tocolytic Effect of NSAIDs



- **Indomethacin** is the most commonly used tocolytic agent in this class

NSAIDs-induced Closure of Ductus Arteriosus

- PGE binds EP_4 receptor to stimulate adenylate cyclase and a subsequent vasodilation of ductus arteriosus

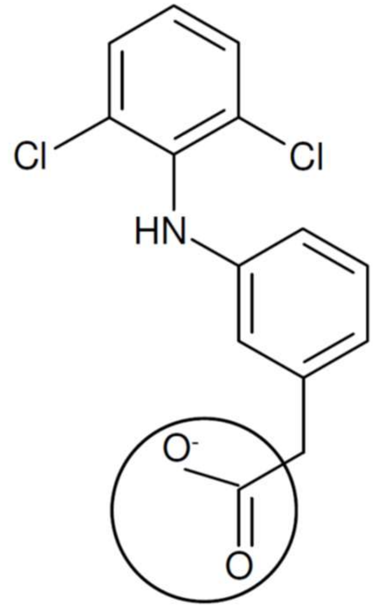


J Clin Invest. 2006;116(11):2863-2865.

Common Pharmacokinetic Characteristics of NSAIDs

- Hydrophobic
- Low pKa (~3-5)
  - Orally active
  - Accumulate at the site of inflammation (pH ~5.5 or lower)
    - Prolonged duration of action relative to their plasma $t_{1/2}$
- Highly bound to plasma protein (95-99%), primarily albumin
- Distributed widely
  - Achieve sufficient concentrations in the CNS to have a central analgesic effect
  - Synovial fluid concentrations ≥ half the plasma concentration
- Excreted by either glomerular filtration or tubular secretion

Diclofenac in the inflamed tissues



Treated rats (Carrageenan injected)	t_{max} (h)	C_{max} (nmol/g)	AUC(0-t_{last}) (h*(nmol/g))	t_{last} (h)	$t_{1/2}$ (h)
Control footpads	1	0.16	0.92	8	1.80
Inflamed footpads	4	1.30	16.00	24	5.30
Inflamed nape neck	8	1.30	20.00	24	7.10

BMC Pharmacology 2009;9:5. doi:10.1186/1471-2210-9-5

Meloxicam in the inflamed tissues

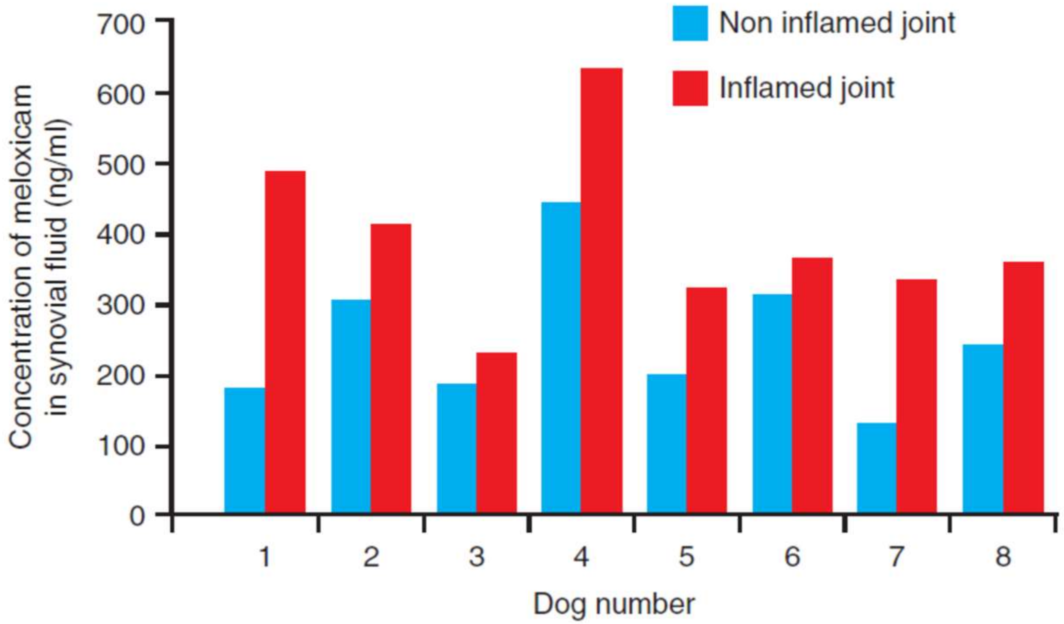


FIG 1: Meloxicam concentration in individual joints

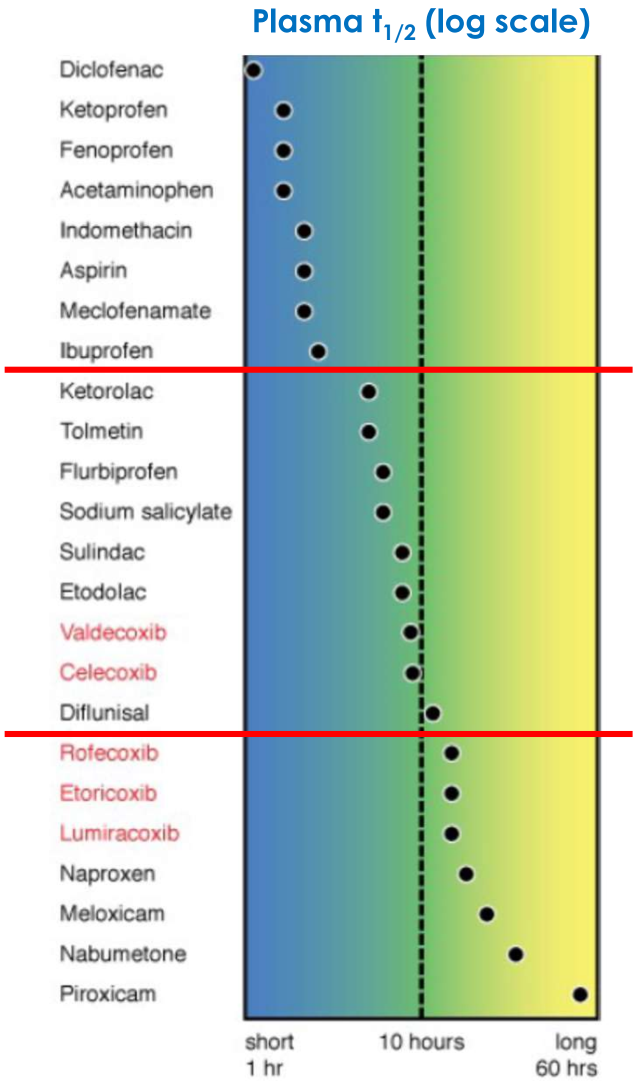
TABLE 1: Mean meloxicam concentration in normal versus inflamed joints

Mean meloxicam concentration in normal joint (ng/ml)	246 (SEM 35.3)
Mean meloxicam concentration in inflamed joint (ng/ml)	391 (SEM 43.4)
P Value	P=0.002

Veterinary Record. 2012:170;207a. doi: 10.1136/vr.100306

Common Pharmacokinetic Characteristics of NSAIDs (Cont.)

- Metabolism
  - Hepatic biotransformation
    - Most of tNSAIDs (multiple CYPs/non-CYPs)
    - Celecoxib (CYP2C9)
    - Etoricoxib (CYP3A4)
    - Parecoxib $\xrightarrow{\text{hydrolysis}}$ Valdecoxib (CYP3A4, 2C9)
  - Several NSAIDs/their metabolites are glucuronidated or conjugated
- In general, NSAIDs are not recommended in severe hepatic or renal disease
  - Adverse effects

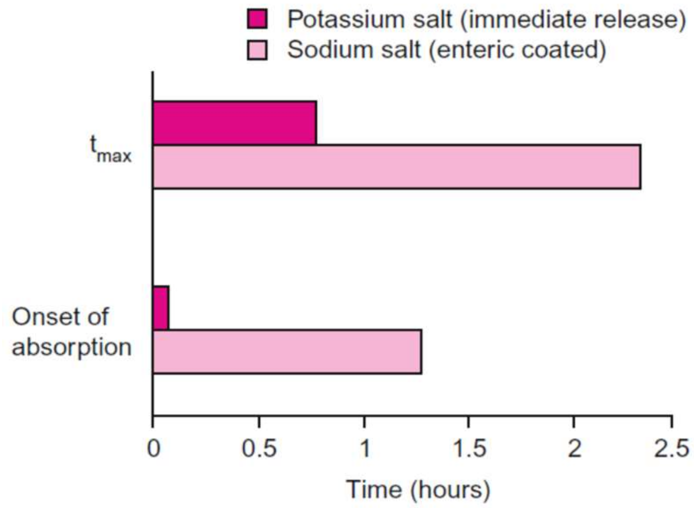


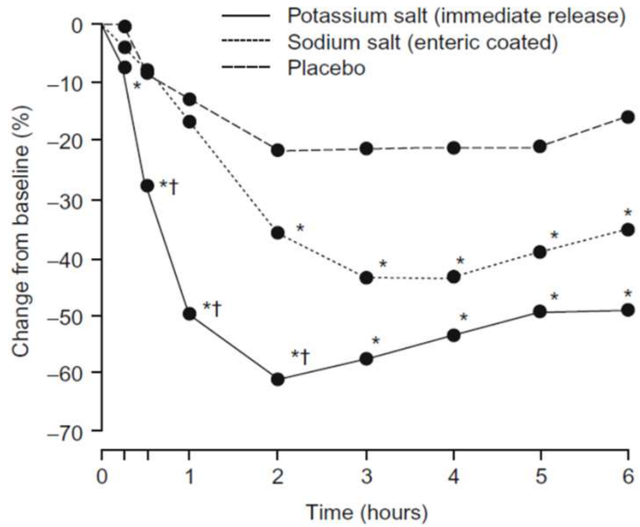
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 12e.

Sodium vs. Potassium Salt of Diclofenac

- **Diclofenac potassium**
  - More water soluble
  - More rapid dissolution
  - Faster absorption

Shorten time to onset of analgesia



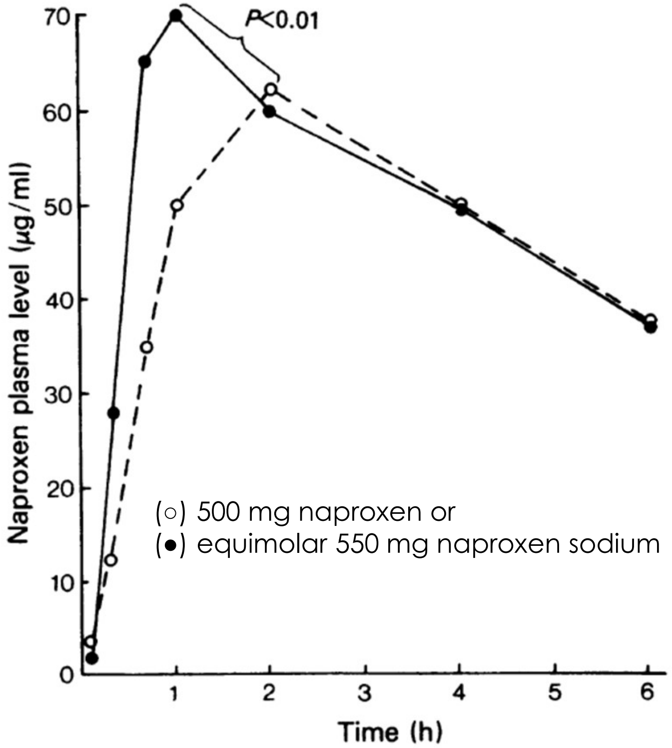


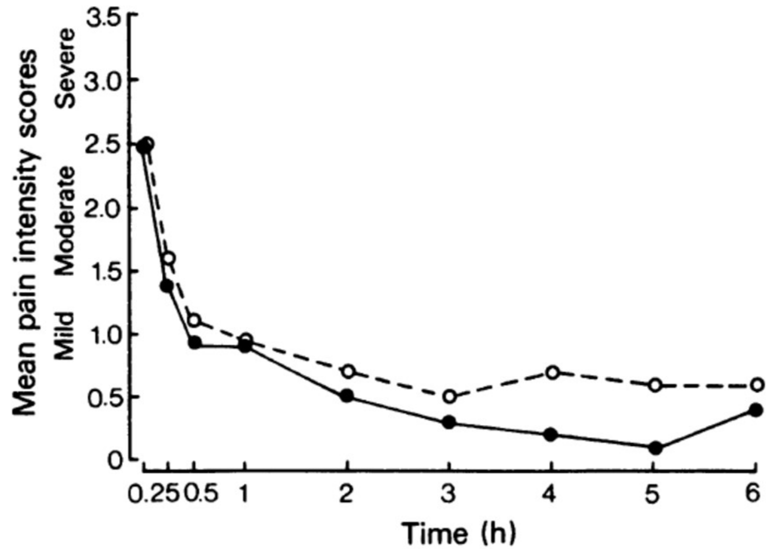
Drugs 2015;75:859–877

Drugs 1999; 57 (6): 991-1003

Naproxen vs. Naproxen Sodium

- Naproxen sodium is more rapidly absorbed from the GI tract
- But, onset of analgesic effect is similar (e.g. in moderate-to-sever uterine cramps)

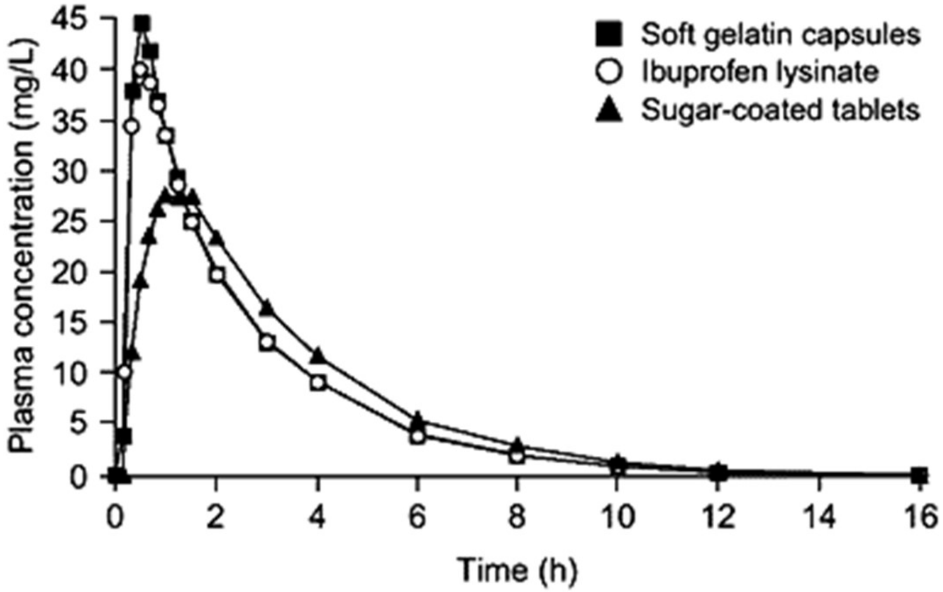




Br J Clin Pharmac 1980;10:259-263

Ibuprofen Tablet vs. Soft Gelatin Capsule

- Soft gelatin capsule has **a faster absorption** and **higher peak serum concentration**
- <u>But, onset of analgesic effect is unclear</u>



Clin Drug Invest. 2001;21(1):73-78.

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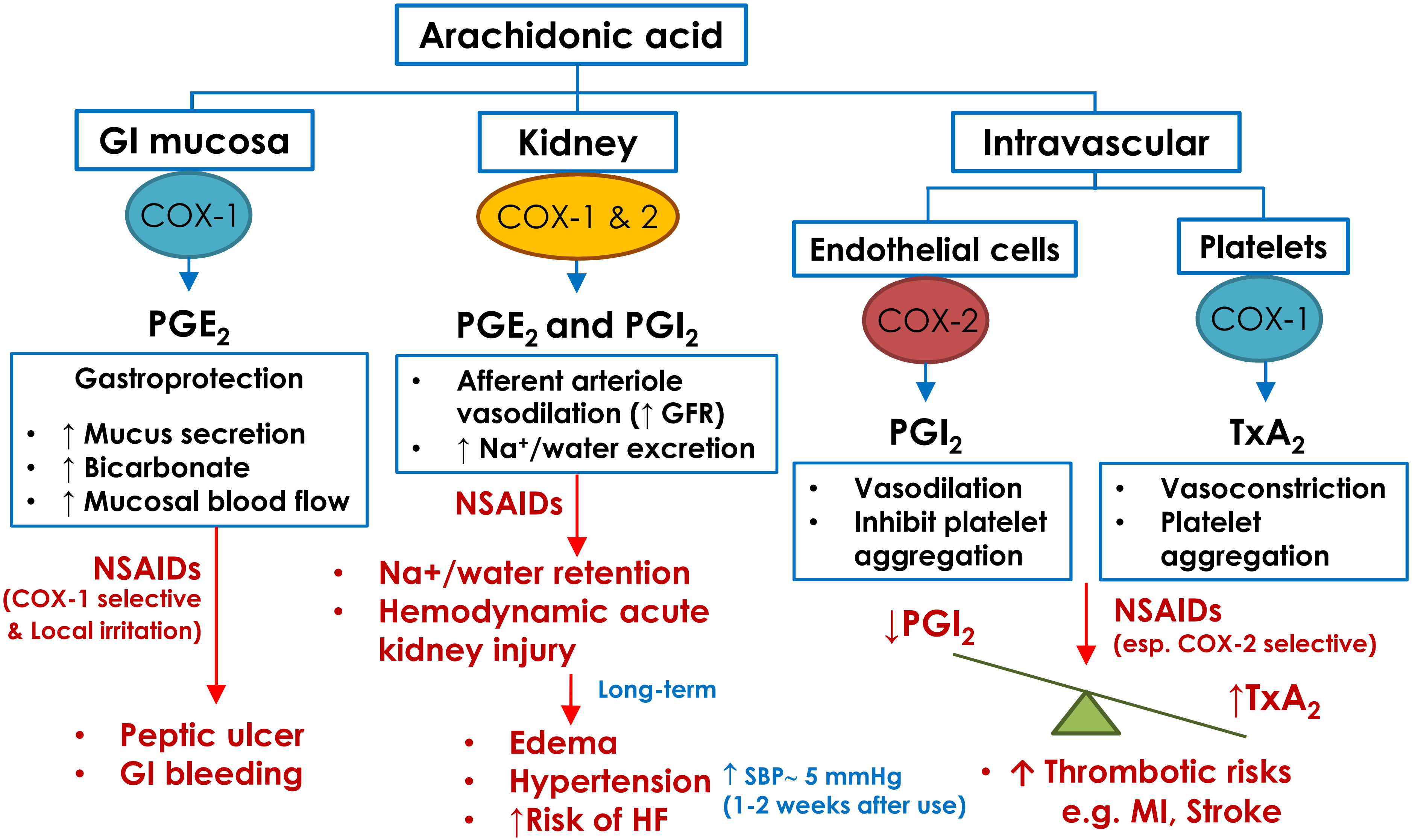
Adverse Effects of NSAIDs



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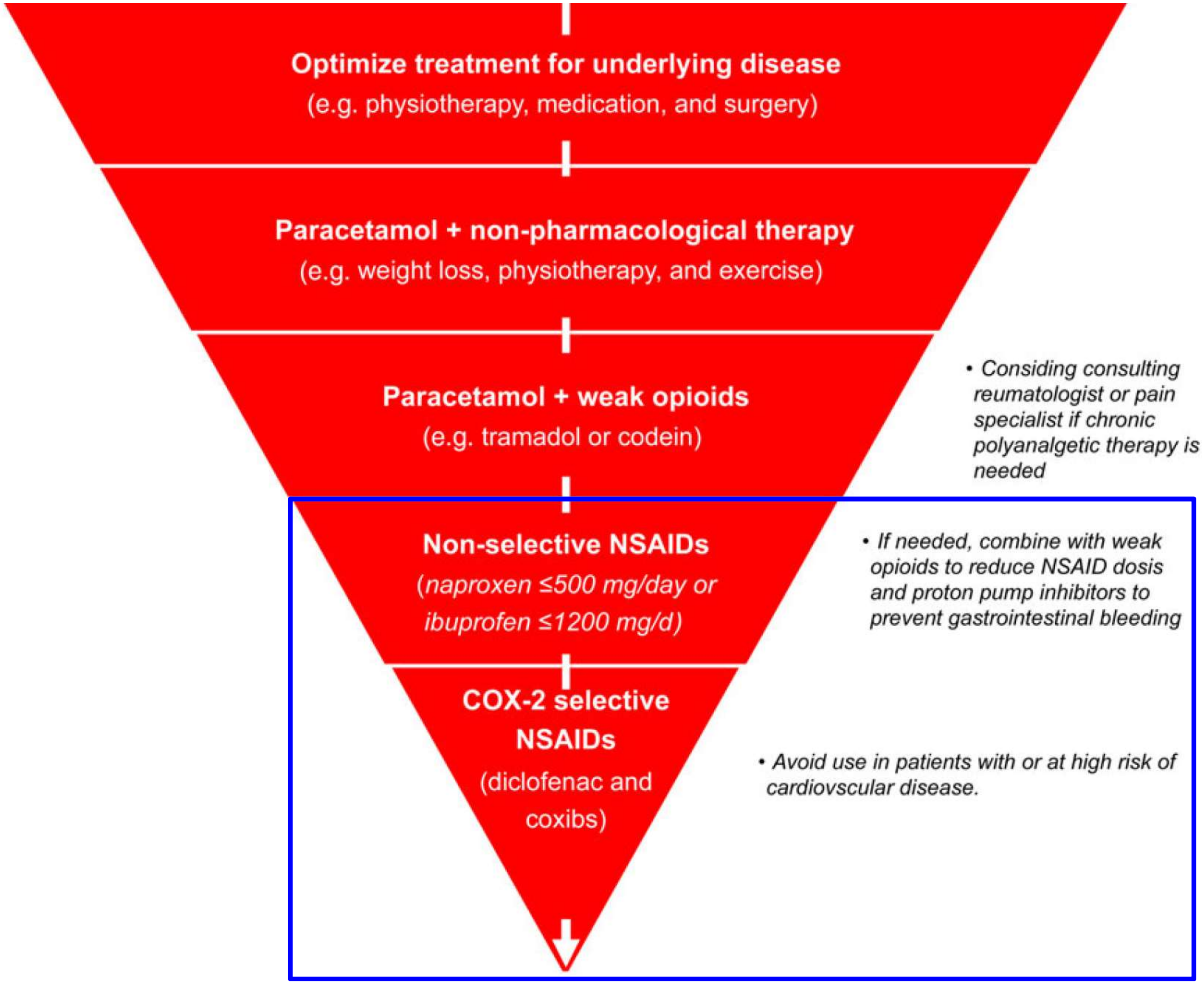
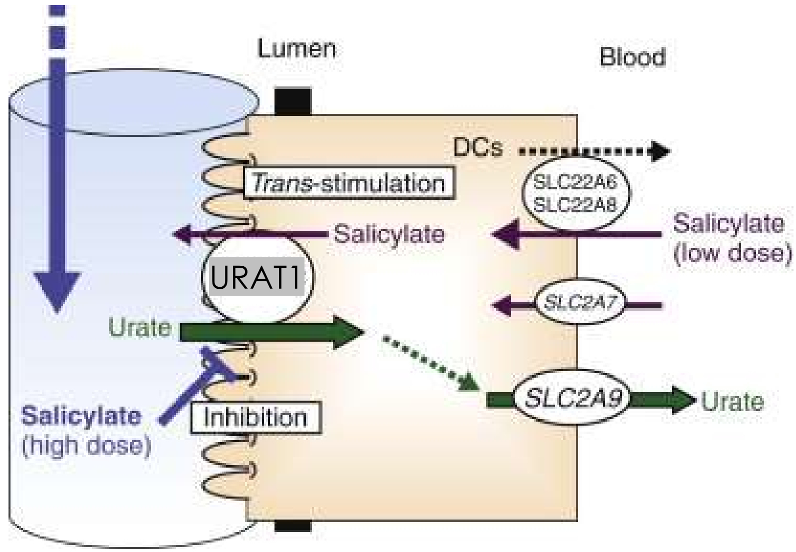
NSAIDs of Choice for CVD patients



European Heart Journal – Cardiovascular Pharmacotherapy (2016) 2, 108–118

Low-dose Aspirin Increases Serum Urate

- **Low-dose aspirin**
  - Stimulate/upregulate urate/anion exchanger 1 (URAT1)
  - Associated with an increased risk of recurrent gout attacks
  - To help avoid the risk of gout attacks associated with low-dose aspirin
    - Serum urate monitoring with concomitant use
    - Dose adjustment of a urate-lowering therapy may be especially important



Gout & Other Crystal Arthropathies 2012:51-58.

Ann Rheum Dis. 2014;73(2): 385–390.

Aspirin Uses and Reye's Syndrome

- Occur in children < 20 year old
  - When use aspirin/salicylate in viral infection-induced fever
- Unclear mechanism
- Symptoms
  - Acute encephalopathy
  - Liver dysfunction
  - Fatty infiltration of liver and other viscera

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NSAIDs Hypersensitivity

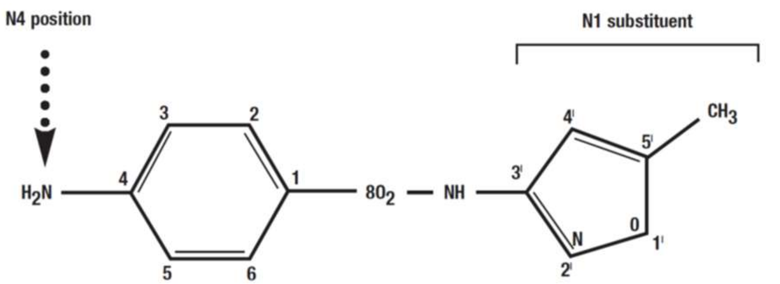
Drug hypersensitivity reactions
(ADR type-B, unpredictable)

- **Non-allergic** (Cross-reactivity)
  - NSAIDs exacerbated respiratory disease — A shift in AA metabolism to leukotrienes may contribute (~10% of tNSAIDs users)
  - NSAIDs exacerbated cutaneous disease
  - NSAIDs induces urticaria-angioedema — Symptoms are induced by at least two NSAIDs with different chemical structure (not belonging to the same chemical group).
- **Allergic** (Selectivity)
  - Single NSAIDs-induced urticaria/angioedema or anaphylaxis — Symptoms are induced by NSAIDs that belong to the same chemical group
  - NSAIDs-induced delayed hypersensitivity reactions

Allergy Asthma Immunol Res. 2015;7(4):312-20.

Cross-reactivity with Sulfonamide Allergy?

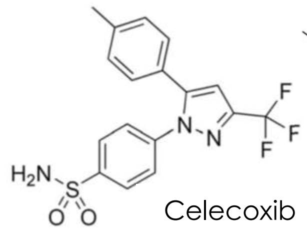
Sulfonamide antibiotics



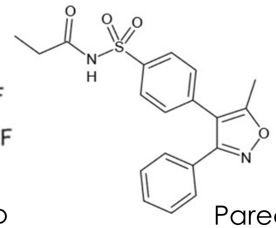
Non-type I, hypersensitivity

Type I, immediate hypersensitivity

Sulfonamide (SO_2NH_2)

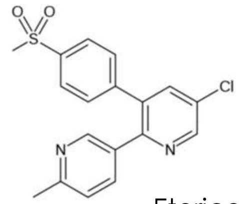


Celecoxib



Parecoxib

Sulfone (SO_2)



Etoricoxib

- Given the weight of available evidence
  - The COX-2 inhibitors: celecoxib, etoricoxib, parecoxib
    - <u>Should not</u> be expected to <u>cross-react</u> with sulfonamide antimicrobials in clinical practice (Am J Health-Syst Pharm. 2013; 70:1483-94.)
    - But, in prescribing information
      - <u>Sulfonamides: Celecoxib and parecoxib are contraindicated</u>
      - Sulfone: Etoricoxib is not contraindicated

Drug Interactions with NSAIDs

- **ACEIs/ARBs**
  - ↓ Hypotensive effect
  - ↑ Hyperkalemia
- **Warfarin**
  - ↑ Risk of Bleeding
- **Steroids and SSRIs**
  - GI complications
- **Lithium**
  - Most of NSAIDs ↓ renal excretion of lithium
  - Sulindac ↓ plasma lithium level

Examples of Clinical Application of NSAIDs

- Treatment of fever (Select NSAIDs with rapid onset)
- Treatment of inflamm. & pain from various origins
  - Pain at low to moderate intensity
    - Mainly somatic pain
      - Headache, including migraine (1^{st} line drug for migraine attack)
      - Musculoskeletal disorders, including OA, RA, gout
      - Post-operative pain, dental pain
    - Some visceral pain (e.g. menstrual pain)
    - Not effective against neuropathic pain

Examples of Clinical Application of NSAIDs (Cont.)

- Prevention and treatment of thromboembolic events (e.g. IHD, Stroke)
  - **Low-dose aspirin**

- Treatment of Kawasaki disease
  - **Aspirin**
    - Initially at high dose: as **anti-inflammatory** and **antipyretic** effects
    - Then (e.g. 2-3 days after afebrile) at low-dose: as **antiplatelet** effect to prevent coronary artery abnormalities (e.g. aneurysm, IHD) for 6-8 wks

- Prevention of preterm labor/induction of PDA closure
  - **Indomethacin**

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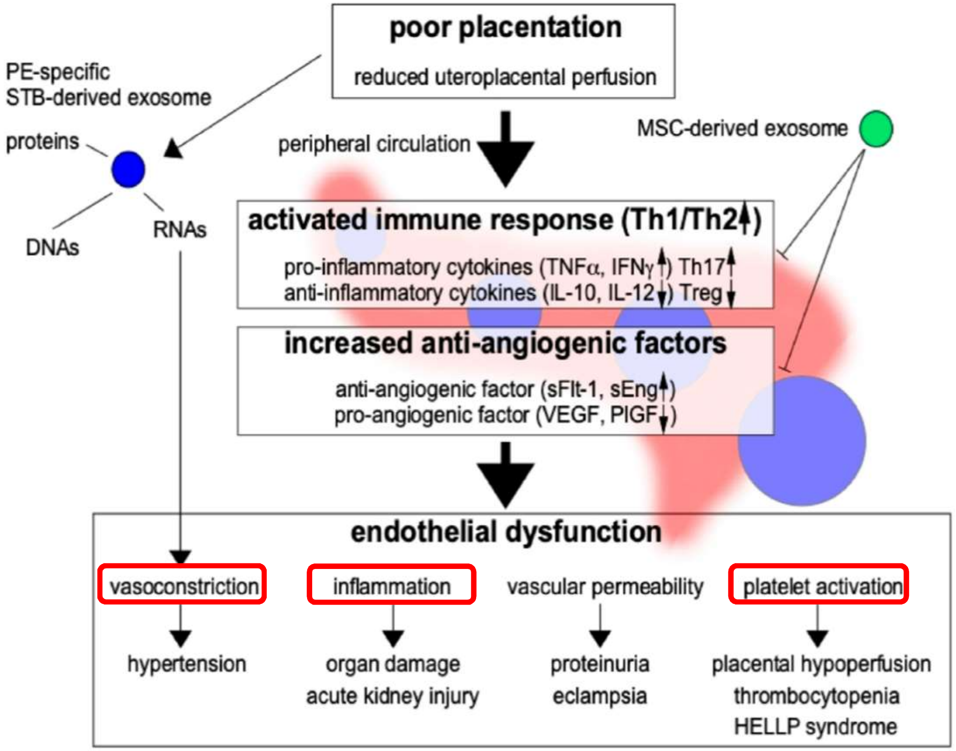
Examples of Clinical Application of NSAIDs (Cont.)

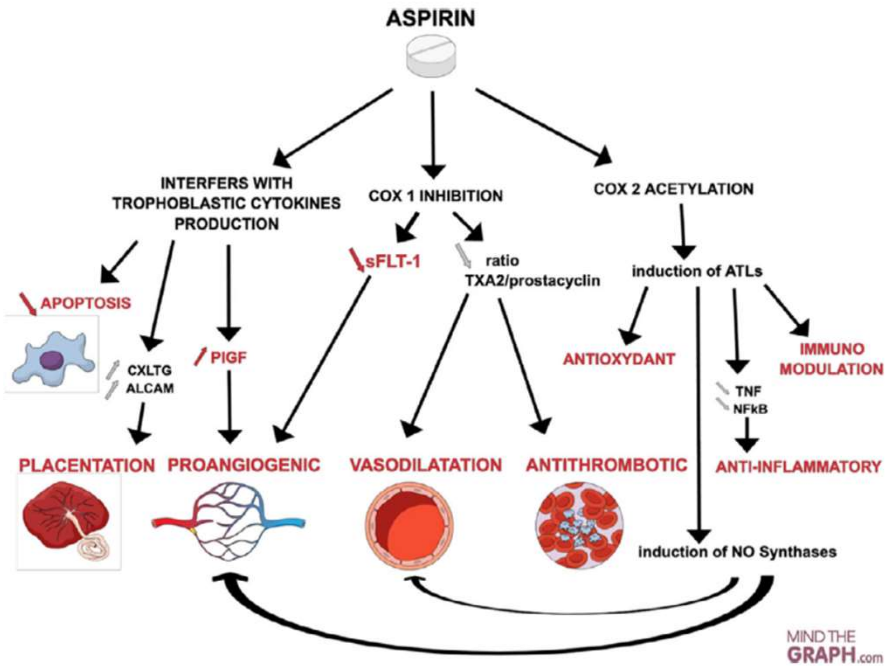
- Prevention of pre-eclampsia in high-risk pregnancy
  - **Low-dose aspirin, initiated before 16 weeks**



Int. J. Mol. Sci.2021;22:2572.

High-risk Factors	Moderate-risk Factors
• Hypertensive disease in previous pregnancy • Chronic kidney disease • Autoimmune disease, eg, SLE and antiphospholipid • Type 1 or type 2 diabetes • Chronic hypertension	• First pregnancy • Age 40 or older • Pregnancy interval >10 y • BMI > 35 at first visit • Family history of PE • Multifetal pregnancy

Abbreviations: BMI, body mass index; SLE, systemic lupus erythematosus.



Prenatal Diagnosis. 2020;40:519–527.

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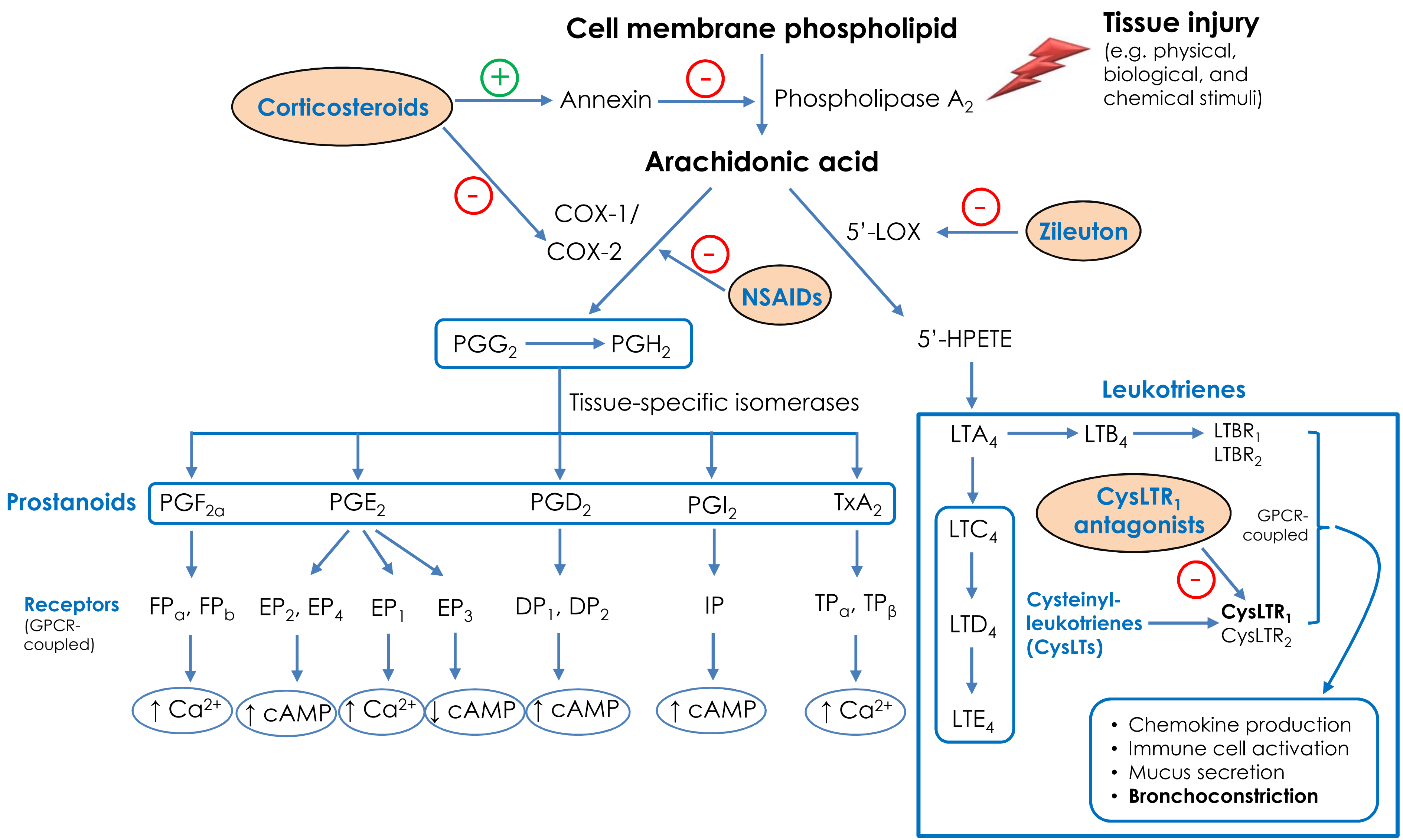
Eicosanoid Synthesis Pathways



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Distribution of $CysLTR_1$ and 5-LOX in The Airways

Expression of LT receptor subtypes, LT biosynthetic pathway enzymes and LT production in different cell types[a]

Cell type	$CysLT_1$ receptor	$CysLT_2$ receptor	BLT_1 receptor	BLT_2 receptor	5-LO	FLAP	LTC_4 Synthase	CysLT production	LTB_4 production
Eosinophils	+	+	–	–	+	+	+	+	+
Basophils	+	+	–	–	+	+	–	+	+
Mast cells	+	+	+	+	+	+	+	+	+
Monocytes	+	–	–	–	+	+	+	+	–
Macrophages	+	–	–	–	+	+	+	+	+
Neutrophils	+	–	–	–	+	+	–	+	+
T lymphocytes	+	–	+	–	–	–	–	+	–
B lymphocytes	+	–	–	–	–	–	–	–	–
Hematopoietic stem cells	+	–	–	–	+	–	–	–	–
Endothelial cells	+	+	–	–	–	–	–	–	–
Glandular cells	+	–	–	–	–	–	–	–	–
Smooth muscle cells	+	–	–	–	–	–	–	–	–
Fibroblasts	+	–	+	–	+	+	+	+	+
Platelets	–	–	–	–	–	–	+	+	+
Erythrocytes	–	–	–	–	–	–	+	+	+
Epithelial cells	–	–	–	–	–	–	+	+	+

[a] +, presence of receptors or enzymes or production; –, absence or lack of evidence of receptors or enzymes or production, to the best of our knowledge.

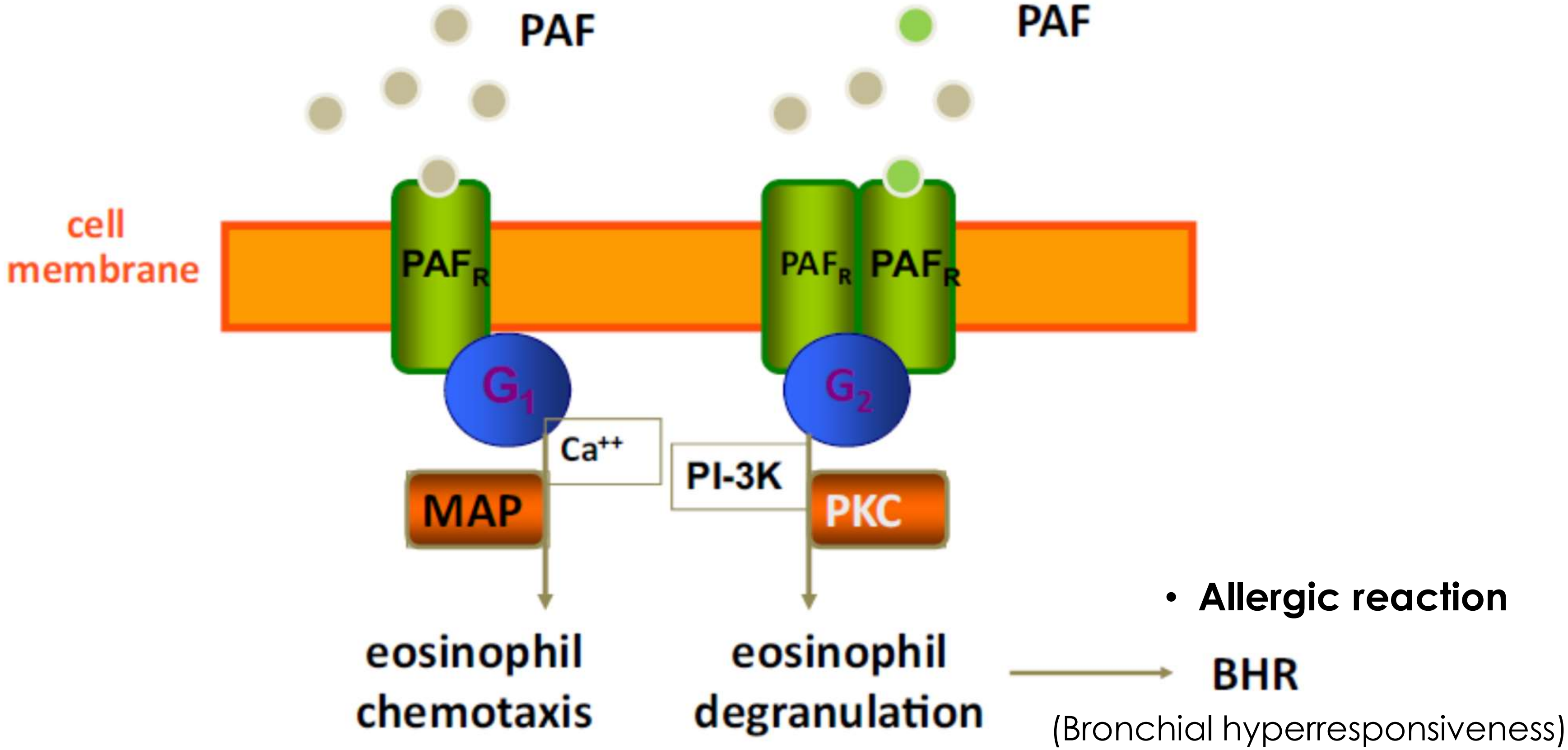
Drug Discov Today. 2007;12(9-10):404-12.

Pharmacology of Anti-leukotrienes

Drug	Montelukast	Pranlukast	Zafirlukast	Zileuton
MOA	$CysLTR_1$ antagonists			5-LOX inhibitor
Pharmacokinetics • $t_{1/2}$	2.7-7 h	3-9 h	10 h	2.5 h
Indications	**Asthma** • Allergen-challenged • Exercise-exacerbated • Aspirin-exacerbated			
	Allergic rhinitis			
Side effects	• Headache • Abdominal pain • Liver enzyme elevations		Possible association with Churg–Strauss syndrome	

[1]Pediatr Drugs 2012: 14 (5): 317-330. [2]Drug Discov Today. 2007;12(9-10):404-12.

Platelet activating factor (PAF) antagonists



- **Allergic reaction**

Allergy 2015;70:1–24.

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Platelet activating factor (PAF)

- PAF is another lipid-derived autacoid

	Characteristic
Source	Mast cells, basophils, eosinophils, monocytes, fixed tissue macrophages, neutrophils, endothelial cells, platelets
Inactivation	By platelet-activating factor acetylhydrolase (PAF-AH) to lyso-PAF
Mean plasma level	23.8 pg/mL
Half-life of PAF	3–13 min

Int J Immunopathol Pharmacol. 2015;28(4):584-9.

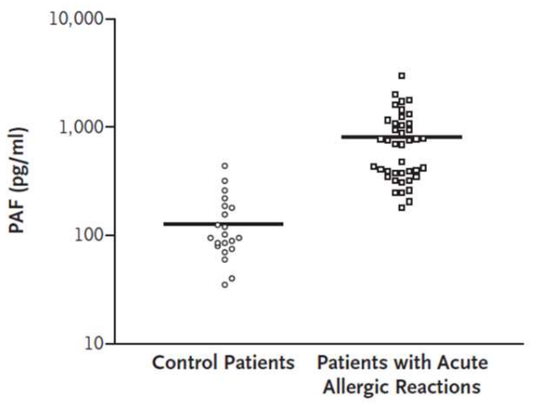

Figure 1. Serum Platelet-Activating Factor (PAF) Levels in Patients with Acute Allergic Reactions.
Untransformed values are plotted on a log scale.

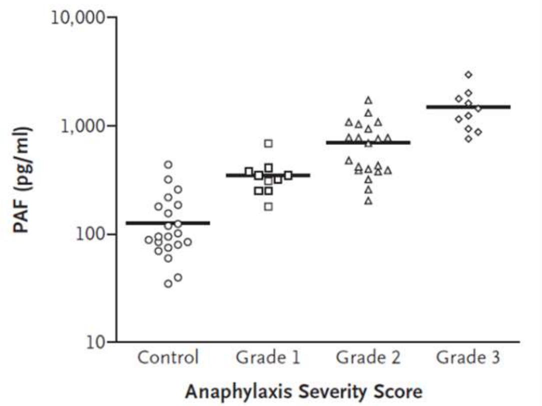

Figure 2. Serum Platelet-Activating Factor (PAF) Levels as a Function of Anaphylaxis Severity Score.
Untransformed values are plotted on a log scale.

N Engl J Med 2008;358:28-35.

Rupatadine

- Inverse agonist of the histamine H_1-receptor
- Specific and dose-dependent inhibition of PAF receptors
- Metabolized by CYP3A4
  - Active metabolites have anti-histamine activity, which contribute to its long duration of action
- Plasma half-life: 4-5 h
- Indications: Allergic rhinitis, Chronic spontaneous urticaria

Adverse event*	Rupatadine-treated patients n = 2025 (%)	Placebo-treated patients n = 1315 (%)
Somnolence	9.5	3.4
Headache	6.8	5.6
Fatigue	3.2	2.0
Asthenia	1.5	0
Dry mouth	1.2	0
Dizziness	1.0	0

*Possibly, probably or definitely related to study medication.

Allergy 2015;70:1–24.

Prostaglandin Analogues/Derivatives

Prostanoids	Indication (mechanism of action)
Prostaglandin E1 • Alprostadil • Lubiprostone • Misoprostol	• Erectile dysfunction (relaxation of corpus cavernosal smooth muscle) • Constipation (activates Cl^- channel) • Mucoprotective (activate mucus secretion in GI) and Oxytocic (myometrial contraction)
Prostaglandin E_2 • Dinoprostone	• Oxytocic
Prostaglandin F_{2a} • Bimatoprost • Latanoprost • Travoprost • Carboprost	• Reduces intraocular pressure in glaucoma (increase outflow of aqueous humor) • Oxytocic
Prostacyclin (PGI_2) • Epoprostenol • Iloprost • Treprostinil	• Pulmonary arterial hypertension (relaxation of pulmonary artery)

Any Questions?

Post-test

Thank You